

Thinking—Fast, Slow, and Artificial

How AI is Reshaping Human Reasoning and the Rise of Cognitive Surrender”

1. Core Thesis: AI as a Third Cognitive System

The paper argues that traditional “dual-process” models of thinking (fast intuition vs. slow deliberation) are incomplete in the age of generative AI. The authors introduce **Tri-System Theory**, adding **System 3: artificial cognition**—external, algorithmic reasoning systems such as large language models. These systems are no longer just tools; they actively participate in decision-making and can reshape how people think.

2. From Assistance to Cognitive Restructuring

Unlike calculators or reference books, modern AI systems generate judgments, explanations, and recommendations that users often adopt directly. The paper contends that AI is changing not just outcomes but the *process architecture* of reasoning. Decision-making increasingly unfolds in a triadic ecology: human intuition (System 1), human deliberation (System 2), and AI outputs (System 3).

3. Defining “Cognitive Surrender”

A central concept is **cognitive surrender**: the tendency to adopt AI outputs with minimal scrutiny, effectively substituting external reasoning for one’s own. This differs from strategic “cognitive offloading” (e.g., using a calculator). In surrender, individuals relinquish oversight and accept AI answers as their own, even when errors are present.

4. Experimental Design (What They Tested)

Across three preregistered experiments (N=1,372; 9,593 trials), participants completed reasoning tasks (Cognitive Reflection Test items) either without AI or with access to an AI assistant. Critically, AI accuracy was experimentally manipulated—sometimes correct, sometimes confidently wrong. Participants retained full autonomy over whether to consult and whether to follow the AI.

5. Core Finding: Accuracy Tracks AI Quality

When AI was accurate, participant accuracy rose sharply (+25 percentage points relative to baseline). When AI was faulty, performance dropped significantly (–15 percentage points). This swing—large and symmetrical—demonstrates cognitive surrender: users’ performance largely mirrored the AI’s correctness rather than their own baseline reasoning ability.

6. Confidence Inflation Risk

Access to AI increased participants’ confidence—even when AI provided incorrect answers. Notably, confidence did not significantly decline after repeated exposure to faulty AI. This combination—performance dependence plus elevated confidence—raises important governance concerns in domains requiring accountability.

7. Time Pressure: AI as Cognitive Buffer

Under time pressure (Study 2), human-only performance deteriorated, as expected from dual-process theory. However, users who engaged AI were buffered from time-related decline—*if* the AI was accurate. When AI was faulty, performance still dropped. Thus, AI can compensate for cognitive load constraints, but only to the extent that its outputs are reliable.

8. Incentives and Feedback: Partial Mitigation

When participants received performance incentives and item-level feedback (Study 3), override rates increased and uncritical adoption decreased. Cognitive surrender was reduced—but not eliminated. This suggests that structural accountability mechanisms (feedback loops, incentives) can reactivate deliberative oversight.

9. Who Is Most Susceptible?

Individual differences mattered. Higher **trust in AI** predicted greater adoption and greater vulnerability to faulty outputs. Lower **Need for Cognition** and lower **fluid intelligence** were associated with more surrender behavior. Importantly, even higher-ability individuals were not immune; they simply overrode faulty AI more often.

10. Strategic Implications for Leaders

For organizations, the implications are structural rather than tactical. AI integration does not simply “improve decisions”—it reconfigures cognitive responsibility. AI can enhance performance under complexity and time pressure, but also create systematic overconfidence and accountability diffusion. To mitigate risks, leaders should:

- Implement feedback and verification mechanisms
- Design workflows that preserve human deliberation for high-stakes decisions
- Monitor overreliance patterns
- Align incentives with verification rather than mere speed

The paper concludes that AI is not merely augmenting cognition—it is becoming a co-agent in reasoning. Managing this triadic cognitive system will be central to maintaining judgment quality, autonomy, and accountability in AI-embedded organizations.